MRL - Milestone Review Log

Part 1 · Milestone plan · Purpose and decision standard

v1.0

Study / protocol	CIP version / date	Site / code	MRL plan ID
Plan revision	**Effective date**	**Principal Investigator**	**Date opened**
Linked SRR ID / revision	**Linked POT plan ID**	**Plan prepared by / role**	**Controlled location of plan and reviews**

A Purpose and use

A **milestone** is a point in the investigation where a failure, if it goes unnoticed, repeats in every case that follows: the first procedures, the immediate postoperative period, the approach to endpoint assessment, close-out. At each milestone the PI takes a **gate decision**: the unit concerned (the site, a surgeon, a process or a cohort) may take its next step, may take it only within stated limits, or must wait. Participants move through the phases at different times, so a gate decides a step for its unit, not a pause for the whole study.

The MRL has two parts. Part 1 is the plan: which milestones this investigation uses, what triggers each one and what must be shown at it, approved by the PI before the first participant, with an index of every gate decision taken. Part 2 records one gate review, in a new copy each time. The MRL cites the records that hold the evidence; it does not repeat them, and its decisions do not close them.

	Part 1 · Milestone plan	Part 2 · Gate review record
Sections	A to F	I to V
Completed	Once, before the first participant. Changes go in D and apply only from their effective date.	At each gate review, in a new copy. V only when a condition or block needs clearing.
By	Record owner; approved by the PI.	Evidence reviewer prepares I to III; the PI decides in IV.
Reads	CIP, SRR and the POT plan.	The records cited as evidence: ICDL, SLCT, PFVR, FUCR, IQL, IMC, SART, CSST, PIR, CAPA log and others.

B Decision standard

Gate decision	Use when	The decision must state
Pass	The trigger was reached for the right unit and denominator, the evidence supports the next step and known limitations do not prevent it.	The scope now permitted. Pass does not mean clinical results are good.
Pass with conditions	The evidence supports the next step only within an explicit limit.	Permitted and prohibited scope; for each condition its owner, destination record, clearing evidence, re-review trigger and what happens if it is not met.
Blocked	The trigger, an essential control, an authorization or decision-critical evidence is missing, insufficient or materially uncertain.	Exactly what cannot proceed and the evidence needed to re-review. A block need not stop the whole study.

C Milestones used in this investigation

Options, not a fixed sequence: use a milestone only when its decision changes what the site may do next, and set it before the evidence exists. A calendar date alone is not a trigger.

ID	Milestone · decision it enables	What to verify · typical records	Use?	Unit	Trigger and denominator	Review window	Permitted next step
M0	**Feasibility** Site capability accepted	Operating, imaging and rehabilitation capacity; imaging gap assessment; long-term follow-up logistics. Feasibility record, SIRR.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M1	**Start-up** Release to enroll	Delegation with criticality codes; surgeons released and device training documented; assessors certified; instruments calibrated; imaging charter agreed; SRR open; IMP approved. RACI, CCDM, DTRM, TRE, SCR, SIRR, SRR, IMP.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M2	**First enrollments** Continue or expand enrollment	Consent, eligibility evidence, dual review where imaging decides, PI resolution of borderline cases, screening log. PSEAR, SIFSL.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M3	**Procedural implementation** Next procedural scope	Release current for each case, implant traceability, critical steps and substitutions documented, due learning-curve review. SCR, TRE, ICDL, SLCT.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M4	**Immediate postoperative period** Pathway continues for the cohort	Immediate review of complications and device performance; postoperative images usable. ICDL, IQL, SART, SCML.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M5	**Approach to endpoint assessment** Endpoint assessment proceeds	Visits in window, PROMs on schedule, images usable, endpoint data agreement. PFVR, FUCR, IQL, IMC, PIR.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M6	**Close-out** Controlled close-out	Device reconciliation signed; data completeness per participant; final safety report; archive. Accountability, IMC, PIR, SART.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M7	**Continuous improvement** Lifecycle closure	Corrective actions closed and effective in their records; lessons in standing procedures. CAPA log, PIDLR.	☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			
M_			☐ Yes ☐ No	☐ Site ☐ Surgeon ☐ Process ☐ Cohort			

M0 does not replace sponsor site selection or institutional approval. M1 is executable readiness, not completed forms. M3 does not certify a completed learning curve (SLCT). M5 is readiness of the assessment process, not final results. M6 leaves open cases and CAPAs in their own records. The safety stream is not a milestone: it is checked in every gate review (Part 2, III).

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Part 1 · Milestone plan · Changes and approval

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D Changes to the plan

A change applies from its effective date; decisions already taken stay with the plan revision they used.

Change ID	Milestone	Previous setting	New setting	Reason	Effective from	Approved by

E Plan approval

The PI approves the plan before the first participant and at each revision.

Plan revision approved		Approval date	

Plan prepared by

Name / role:

Signature / date:

Principal Investigator

Name:

Signature / date:

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Part 1 · Milestone plan · Gate decision index

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Study / site	MRL plan ID	Index sheet no. / total	Last entry date

F Gate decision index

One line per signed gate decision, including re-reviews, so the sequence can be read at a glance. The detail stays in the Part 2 copy with the same Review ID.

Review ID	Milestone	Unit	Trigger met (date)	Decision date	Decision	Scope now permitted	Next milestone or re-review trigger
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		
					☐ Pass ☐ With conditions ☐ Blocked		

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Part 2 · Gate review record · Identification and evidence

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Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total

I Gate identification

Milestone (Part 1, C)		**Implementation unit**	
Trigger met · date		**Count reached · denominator**	
Evidence period · cut-off		**Next step requested**	
Previous decision (re-review)		**Prepared by · date**	

II Evidence package

Cite each record so it can be found again: instrument, record ID, version or date, case or period covered, location. Adequate describes the item; it does not decide the gate.

Item	Control or element	Source · record ID · date · location	Coverage	Status	What it shows for this gate
E-01				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	
E-02				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	
E-03				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	
E-04				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	
E-05				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	
E-06				☐ Adequate ☐ Partial ☐ Unavailable ☐ N/A	

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Part 2 · Gate review record · Sufficiency

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Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total

III Sufficiency for this gate

For this milestone and unit only: judge whether the evidence in II is enough for the decision. Do not repeat the review done by the IMC or the specialist record.

1 · Trigger and scope

ID	Question	Result	Evidence item	Note
S-01	The configured trigger was reached for the right unit and the actual denominator.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		
S-02	The population, evidence period and requested next step match the plan.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		

2 · Controls and evidence

ID	Question	Result	Evidence item	Note
S-03	The critical controls for this gate were in place and their use is evidenced.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		
S-04	Release, authorization, qualification and supervision are current in their records.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		
S-05	Source records and internal verification are retrievable and enough to reconstruct what was done.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		
S-06	Required safety records, due follow-up and the latest trend review are current.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		

3 · Sufficiency

ID	Question	Result	Evidence item	Note
S-07	The evidence supports the next step with no material unknown impact from a critical gap.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		
S-08	Open issues have an accountable route and their limits fit the proposed scope.	☐ Complete ☐ Partial ☐ Not shown ☐ N/A		

Decision-limiting gap or unknown impact

(if none, say so and cite the basis)

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Part 2 · Gate review record · PI decision

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Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total

IV PI gate decision

Choose one decision and state what the team may now do. The decision does not close safety, risk, CAPA or authorization records.

Gate decision	☐ Pass ☐ Pass with conditions ☐ Blocked
Scope permitted from now on	
Scope still not permitted	
PI rationale (cite II and III items)	
Next milestone	☐ Open ☐ Pending ☐ Blocked Objective trigger:

Conditions

Condition ID	Restriction	Destination record · owner	Evidence that clears it	Re-review trigger · latest date	If not met

Principal Investigator · decision

Name:

Signature / date:

Evidence reviewer

Name / role:

Signature / date:

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Part 2 · Gate review record · Condition clearance

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Study / site	MRL plan ID / revision used	Review ID	Sheet no. / total

V Condition clearance and re-review

Only for a condition or block. Cite the evidence actually reviewed. Clearing an MRL condition does not close its destination record; if the permitted scope changes, sign a new Part 2 decision.

Condition ID	Destination record	Evidence reviewed · period	Result	PI conclusion / new Review ID	PI initials · date
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		
			☐ Cleared ☐ Partly cleared ☐ Not cleared		